

No Infinite-Dimensional Unitary Is Weakly l -Sequentially Supercyclic

Candidate full-result packet

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Status. Candidate full answer, subject to expert review, to the strengthened Hilbert-space question in the source. No unitary on a Hilbert space of dimension greater than one is weakly l -sequentially supercyclic. It follows that every weakly l -sequentially supercyclic contraction on an infinite-dimensional Hilbert space is weakly stable.

1 Source question

Kubrusly–Vieira ask whether weak supercyclicity implies weak stability for power-bounded operators, and explicitly sharpen the Hilbert-space version to:

“Does there exist a weakly unstable and weakly l -sequentially supercyclic unitary operator?”

Lemma 3.1 is naturally extended to normed spaces by replacing inner product with dual pairs, and Hilbert-space adjoints with normed-space adjoints.

4. WEAK SUPERCYCLOCITY AND WEAK STABILITY

It was shown in [1 Theorem 2.2] that *if a power bounded operator on a Banach space is supercyclic, then it is strongly stable*. Such a result naturally prompts the question: *does weak supercyclicity imply weak stability for power bounded operators?* The question was posed and investigated in [9], and remains unanswered even if Banach-space power bounded operators are restricted to Hilbert-space contractions, where the problem is equivalently stated for unitary operators (see Remark 4.1 below), and also if weak supercyclicity is strengthened to weak l -sequential supercyclicity: *does there exist a weakly unstable and weakly l -sequentially supercyclic unitary operator?*

Figure 1: The broad and strengthened questions in the source PDF, page 5.

A vector y is weakly l -sequentially supercyclic for U if every x is the weak limit of a sequence $a_k U^{n_k} y$, where n_k is strictly increasing. This is stronger than weak density of the projective orbit.

2 Proof Intuition

If a unitary U had such a vector y , then y would be cyclic. In the cyclic spectral model, U is multiplication by z on $L^2(\mu)$ and y is the constant function 1. A weakly convergent sequence $a_k z^{n_k}$ is norm bounded, so the scalars a_k are uniformly bounded. Its weak limit must therefore be essentially bounded. But an infinite-dimensional $L^2(\mu)$ always contains unbounded functions. This prevents the weak limit set of one projective orbit from being all of the space.

The distinction is exactly what the Kronecker-set example of Bayart–Matheron exploits: its projective orbit is weakly dense, but weak neighbourhoods may require elements of arbitrarily large norm. Weak density does not provide a single weakly convergent sequence.

3 The obstruction

Lemma 1. *Let μ be a probability measure and $C < \infty$. The set*

$$K_C = \{f \in L^2(\mu) : |f| \leq C \text{ } \mu\text{-a.e.}\}$$

is weakly closed in $L^2(\mu)$. If $L^2(\mu)$ is infinite-dimensional, then $L^\infty(\mu)$ is a proper subset of $L^2(\mu)$.

Proof. K_C is convex. It is norm closed: from any norm-convergent sequence one may extract an almost-everywhere convergent subsequence, which preserves the pointwise bound. A convex norm-closed subset of a Banach space is weakly closed.

For the second assertion, if every L^2 function were essentially bounded, the closed graph theorem would make the inclusion $L^2(\mu) \hookrightarrow L^\infty(\mu)$ bounded. Thus $\|f\|_\infty \leq M\|f\|_2$. Applied to indicators, every positive-measure set would have measure at least M^{-2} . The measure algebra would then have only finitely many atoms and $L^2(\mu)$ would be finite-dimensional, a contradiction. $\square$

Theorem 1. *No unitary operator on a complex Hilbert space of dimension greater than one is weakly l -sequentially supercyclic.*

Proof. First suppose the Hilbert space H is infinite-dimensional. If y were a weakly l -sequentially supercyclic vector, the norm-closed linear span of $\{U^n y : n \geq 0\}$ would be all of H : that closed subspace is weakly closed and contains every weak sequential limit of the projective orbit. Consequently y is cyclic.

The cyclic spectral theorem gives a probability measure μ on $\mathbb{T}$ and a unitary map $W : H \rightarrow L^2(\mu)$ such that

$$Wy = 1, \quad WUW^{-1} = M_z.$$

Choose $g \in L^2(\mu) \setminus L^\infty(\mu)$ using Lemma 1. Weak l -sequential supercyclicity would give scalars a_k and increasing integers n_k such that

$$a_k z^{n_k} \rightharpoonup g \text{ in } L^2(\mu).$$

Every weakly convergent sequence in a Banach space is norm bounded. Since $\|a_k z^{n_k}\|_2 = |a_k|$, there is a finite C with $|a_k| \leq C$ for all k . Hence every term belongs to K_C , and weak closedness of K_C forces $g \in K_C \subset L^\infty(\mu)$, a contradiction.

If $2 \leq \dim H < \infty$, diagonalize U . Relative to an eigenbasis, the absolute values of the coordinates of $U^n y / \|y\|$ are independent of n . Thus the normalized projective orbit cannot be dense in the unit sphere, so it cannot be weakly l -sequentially supercyclic. (In finite dimensions weak and norm convergence agree.) $\square$

Corollary 1. *Every weakly l -sequentially supercyclic contraction on an infinite-dimensional Hilbert space is weakly stable.*

Proof. Use the Sz.-Nagy–Foiaş–Langer decomposition quoted in Remark 4.1 of the source:

$$H = H_c \oplus H_u, \quad T = C \oplus U,$$

where C is completely nonunitary and weakly stable, while U is unitary. Let $y = y_c \oplus y_u$ be weakly l -sequentially supercyclic. Projection of approximating sequences onto H_u shows that, if $\dim H_u > 1$, then y_u is weakly l -sequentially supercyclic for U , contradicting Theorem 1.

If $\dim H_u = 1$, write $U = \lambda I$ and choose nonzero $x \in H_c$. To converge weakly to $x \oplus 1$, a sequence

$$a_k (C^{n_k} y_c \oplus \lambda^{n_k} y_u)$$

must have $a_k \lambda^{n_k} y_u \rightarrow 1$, so (a_k) is bounded. Weak stability of C then gives $a_k C^{n_k} y_c \rightharpoonup 0$, contradicting convergence to x . Hence $H_u = \{0\}$ and $T = C$ is weakly stable. $\square$

4 Verification, literature audit, and limitations

The proof is analytic and uses no numerical computation. Its two audit points are: (i) weak convergence of a sequence implies norm boundedness; and (ii) the uniformly L^∞ -bounded ball is weakly closed in L^2 .

A bounded search through 13 August 2026 checked the run indexes, the exact question, the 2024 survey arXiv:2405.02752, the 2023 paper arXiv:2306.08197, the Bayart–Matheron construction, and searches combining “weakly l -sequentially supercyclic”, “unitary”, and “weak stability”. No version of Theorem 1 was found. Several later papers instead describe the Bayart–Matheron unitary as weakly l -sequentially supercyclic. The displayed argument shows why weak density of that orbit cannot imply this stronger direct-sequence property. This is not a claim of priority and the conflict makes independent review essential.

The source’s broader question for merely weak supercyclic operators is not answered positively by this theorem. In fact, the same Kronecker-set construction gives a literature-implied negative answer to that broader question; it is recorded in a companion packet.

Human-review recommendation. Verify the cyclic spectral representation with $y \mapsto 1$, the weak closedness of K_C , and the one-dimensional unitary-summand argument in Corollary 1. Then compare the direct-sequence definition in the source with the weak-density argument in Bayart–Matheron Example 3.6.

References

- [1] C. S. Kubrusly and P. C. M. Vieira, *Boundedly Spaced Subsequences and Weak Dynamics*, Journal of Function Spaces (2018), Article ID 4732836; [arXiv:2010.13331](#).
- [2] F. Bayart and E. Matheron, *Hyponormal operators, weighted shifts and weak forms of supercyclicity*, Proc. Edinburgh Math. Soc. 49 (2006), 1–15, doi:10.1017/S0013091504000975.